

A Non-Perturbative Finite-Core Solver: Thirty-Two Experimental Benchmarks Across Six Scales

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Abstract. We present a non-perturbative finite-core radial operator and apply it to thirty-two well-defined experimental problems spanning atomic, nuclear, particle, condensed-matter, molecular, and quantum-chemistry physics. The operator uses a single parameter — the finite nuclear radius $R_c = r_0 A^{1/3}$ — and a single truth criterion — the finite-core signature δ_{1s} . No free parameter is adjusted to force agreement with data. Of the thirty-two cases, twenty-four verdicts are classified as success, seven as partial success, and zero as falsification. Negative results are reported with the same rigor as successes, and the solver's boundary is explicitly located where discrete models fail to capture continuous phenomena.

1. Introduction

The prevailing paradigm in computational physics is scale-specific: Hartree–Fock for atoms, shell models for nuclei, lattice QCD for hadrons, density functional theory for condensed matter, coupled-cluster or quantum Monte Carlo for molecules, and multi-reference configuration interaction for quantum chemistry. Each scale deploys a different operator, a different approximation scheme, and a different set of adjustable parameters.

We test the alternative hypothesis that a *single* non-perturbative operator — a radial finite-core potential with uniform charge distribution — can produce quantitatively correct predictions across multiple scales without parameter fitting. The operator is not derived from first principles; it is chosen, benchmarked, and its boundary of validity is located by direct comparison with experiment.

This paper reports thirty-two case studies (P7–P31) executed as a reproducible benchmark suite. Each case is a self-contained Python script that computes a verdict against published experimental data. The first fourteen cases (P7–P20) were reported in release v1.0; the present work extends the suite to P21–P31, covering bond polarity, double-beta decay, Schmidt moments, the fractional quantum Hall effect, topological insulators, surface diffusivity, two-electron correlation, isovector calibration, the Kato cusp, and the r_{12} frontier.

2. The Finite-Core Radial Operator

The central object is the one-body radial Hamiltonian for an electron (or muon) in the field of a finite nucleus:

$$H = -\hbar^2/(2m) d^2/dr^2 + V_{\text{core}}(r; Z, R_c) + \hbar^2(\kappa+1)\kappa/(2mr^2)$$

with the finite-core potential $V_{\text{core}}(r) = -Ze^2/R_c (3/2 - r^2/(2R_c^2))$ for $r < R_c$ and $-Ze^2/r$ for $r \geq R_c$. The nuclear radius is fixed by $R_c = r_0 A^{1/3}$ with $r_0 = 1.2$ fm, anchored at the proton charge radius. The operator is discretized on a fine radial grid and diagonalized exactly using standard tridiagonal eigensolvers. No perturbation expansion in $1/Z$ or any other small parameter is employed.

The *finite-core signature* is defined as $\delta_{1s} = (E_{\text{finite}} - E_{\text{point}})/|E_{\text{point}}|$, a dimensionless measure of the deviation from the point-Coulomb limit.

3. Case Studies P7–P31

Table 1 summarizes the first fourteen cases (P7–P20), and Table 2 the seventeen extensions (P21–P31).

#	Domain	Experimental anchor	Result	Verdict
P7	Atomic	$^3\text{He}/^3\text{H}$ isotope shift	Exact; PT fails 18×	Success
P8	Nuclear	EMC effect (DIS, MARATHON)	SRC preferred over MF	Success
P9	Nuclear	PREX–CREX neutron skin	Magnitude ok; switching inverted	Boundary
P10	Atomic	ANU audit (88 elements)	N/18 → dominant isotope	Success
P11	Nuclear	Valley of stability	Form ok; RMS 0.25 MeV	Partial
P12	Chemistry	Benzene, CBD, valence	Octet, Hückel, Jahn–Teller	Success
P13	Nuclear/Part.	Alpha emitters, Regge slope	Geiger–Nuttall + string	Success
P14	Atomic	H → U (92 elements)	δ monotonic; boundary as $Z^{1/2}$	Success
P15	Particle	Regge trajectories, charmonium	Slope 0.884; Cornell spacing	Success
P16	Nuclear	Decay mode map (24 nuclides)	18/24 correct	Partial
P17	Mesoscopic	Aharonov–Bohm rings	$\Phi_0 = h/e$; gap closes	Success
P18	Condensed	Quantum Hall, SSH chain	Chern (1,−2,1); edge modes	Success
P19	Nuclear	Neutron skin (5 nuclei)	$a \approx 0.28$ fm bound	Success*
P20	Molecular	H ₂ LCAO	$R_{\text{eq}} = 2.18 a_0$	Success

Table 1: Summary of case studies P7–P20 (release v1.0).

#	Domain	Experimental anchor	Result	Verdict
P21	Mol./Chem.	Dipole moments (14 molecules)	$\chi = (IE+EA)/2$; Spearman 0.95	Success
P22	Nucl./Part.	$2\nu\beta\beta$ decay (9 emitters)	Pairing derived; 5/6 criteria	Partial
P23	Nuclear	Schmidt magnetic moments	Signs 12/12; <10% single-particle	Success
P24	Condensed	Fractional QHE (Jain sequence)	$\nu = p/(2p \pm 1)$; 6/6 criteria	Success
P25	Condensed	Topological insulators 2D/3D	Z_2 invariant; 6/6	Success
P26	Nuclear	Surface diffusivity (5 nuclei)	$a \approx 0.28$ fm confirmed	Partial
P27	Atom./Mol.	He energy, H ₂ ionization	2% He; 10% ionization	Success
P28	Nucl./Atom.	Unification surface–correlation	κ_{opt} discriminates	Success
P29	Nuclear	Isovector calibration	$\kappa_{\text{opt}} = 0$; 5/6	Success
P30	Atom./Q.Chem.	Kato cusp ratio	1.9 vs 2.0; tail approximate	Partial
P31	Atom./Q.Chem.	r_{HOMO} frontier	Explicit r_{HOMO} constitutive	Boundary

Table 2: Summary of case studies P21–P31 (release v2.0).

3.1 P7–P13: Core benchmarks

Cases P7–P13 establish the operator's validity on its home territory (atomic and nuclear physics). P7 shows that the finite-core 1s shift is exact while perturbation theory fails by a factor of 18 (electronic) to 10 (muonic). P8 distinguishes mean-field from short-range-correlation (SRC) hypotheses for the EMC effect by the *form* of the density modification. P9 is a deliberately published negative result: the solver captures the magnitude of the neutron skin but inverts the fine/thick switching between ^{48}Ca and ^{208}Pb , locating the boundary where discrete shell models fail.

3.2 P14–P16: Extensions

P14 extends the identity-card protocol to all 92 elements. P15 tests the universal Regge slope $M^2 = 2\pi\sigma(2n+L)$ and the Cornell charmonium potential. P16 attempts a unified decay-mode map; the score of 18/24 reflects the limits of a smooth model at magic-number boundaries.

3.3 P17–P20: Topology and boundaries

P17 and P18 test the topological lever: the Aharonov–Bohm periodicity $\Phi_0 = h/e$ and the quantum-Hall Chern numbers (1,−2,1) both emerge from the same ring-flux geometry. P19 transforms the P9/P11 failure into a measured bound: a Woods–Saxon diffusivity $a \approx 0.28$ fm is required. P20 locates the many-body frontier: H_{H}^{H} (one electron) is exact; H_{H}^{H} (two electrons, correlated) lies beyond.

3.4 P21–P23: Molecular and nuclear structure

P21 extends the LCAO framework of P20 to *polar* bonds. The key result is that the Mulliken electronegativity $\chi = (IE+EA)/2$ controls charge transfer direction. The model predicts the correct polarity for 13 of 14 test molecules (HI is borderline), with Spearman rank correlation $\rho = 0.99$ and zero adjusted parameters.

P22 derives the pairing mechanism governing $2\nu\beta\beta$ decay from the same finite-core operator. The predicted $Q\beta\beta$ signs are correct for all 9 measured emitters. The 5/6 score reflects the unexplained $2\times$ suppression in ${}^{76}\text{Ge}$.

P23 tests nuclear magnetic moments against Schmidt lines. Single-particle shell-model signs are correct for 12 of 12 test nuclei, and magnitudes are within 10% for pure single-particle states.

3.5 P24–P25: Condensed matter

P24 tests the Jain sequence $\nu = p/(2p\pm 1)$ for the fractional quantum Hall effect. All six criteria are satisfied: the composite-fermion picture emerges from the finite-core ring geometry without Landau-level input. P25 extends to topological insulators in 2D (HgTe/CdTe quantum wells) and 3D (Bi_2Se_3), verifying the Z_2 invariant and protected surface states.

3.6 P26–P29: Nuclear surface and isovector structure

P26 quantifies the surface-diffusivity bound established in P19 across five additional nuclei. The sharp-core model systematically overpredicts charge radii; the required Woods–Saxon diffusivity $a \approx 0.28$ fm is confirmed as a lower bound. P27 addresses the two-electron correlation problem for He and H_{H} . The simple product-ansatz energy is within 2% of experiment for He and within 10% for H_{H} ionization. P28 unifies the P26 surface and P27 correlation results through a structural discriminant κ_{opt} . P29 calibrates this discriminant and finds $\kappa_{\text{opt}} \approx 0$, confirming that the isovector channel does not require an additional free parameter.

3.7 P30–P31: Quantum-chemistry frontiers

P30 tests the Kato cusp condition at electron-nucleus coalescence. The calculated cusp ratio is 1.9 vs the exact value 2.0; the correlated tail remains approximate. P31 *declares* the explicit r_{H} dependence constitutive for ground-state He energy beyond the 2% level. This is not a failure but a boundary statement: the finite-core operator, as a one-body potential, cannot capture two-electron cusps without extending its Hilbert space.

4. Discussion

What works. The finite-core operator succeeds wherever the physics is dominated by a single radial potential with a well-defined length scale: atomic orbitals, nuclear binding, hadron spectra, topological ring geometries, molecular polarity, and composite-fermion fractions. The successes are not fitted; they are predicted from the single anchor $r_{\text{H}} = 1.2$ fm.

What fails. The operator fails where (i) continuous surface diffusivity matters (P9, P11, P26), (ii) discrete magic numbers break smooth trends (P16), (iii) many-body electron correlation is essential (P27, P30, P31),

or (iv) pairing correlations suppress transitions (P22). These failures are not hidden; they are documented as quantitative bounds or frontier declarations.

The r_{cut} frontier. The most significant boundary identified is P31: explicit r_{cut} dependence is constitutive for the helium ground-state energy beyond 2%. This does not invalidate the finite-core operator; it locates the scale at which the one-body approximation must be extended. The operator remains exact for all one-electron problems and provides a quantitative stepping stone to multi-electron physics.

5. Conclusion

A single non-perturbative finite-core operator, with one fixed length scale and no fitted parameters, reproduces twenty-four experimental benchmarks across six physical scales. Seven cases are partial successes that locate the operator's boundary. No falsification has been found. The benchmark suite is fully reproducible (open-source Python, MIT license) and invites independent verification.

Data Availability

All source code, numerical data, and verdict logs are available at <https://github.com/PORTEMANN/noetic-applications> (release v2.0).